

THE EXPONENT $3/4$ FOR THE EXCEPTIONAL SET OF m/n , IN SHORT INTERVALS AND PROGRESSIONS

ANONYMOUS

ABSTRACT. For an integer $m \geq 4$ call n *m-exceptional* if m/n is not a sum of three unit fractions. We show that there are absolute constants $c, C > 0$ such that every interval I of length $H \geq 2$ contains at most

$$C H \exp\{-c(\log H)^{3/4} m^{-1/4}\}$$

m -exceptional integers, uniformly in $m \geq 4$ and in the position of I . We also give a version for arithmetic progressions, whose only loss is the smooth part of the modulus; a bound for exceptional primes in short intervals that uses no information about primes at the scale of the interval; and a density statement locating the transition from “most primes are m -exceptional” to “most primes are m -representable” at $\log n = m^{1/3+o(1)}$, up to a factor $(\log m)^2(m/\varphi(m))^{1/3}$ in $\log n$ (the “exceptional” side comes from the proof of a theorem of Pomerance and Weingartner). For comparison, Pomerance and Weingartner proved the exponent $2/3$ with $\varphi(m)^{-1/3}$ over initial segments. The method is the one of an earlier internal note on $4/n$ over $[1, N]$, which we use as a black box. What we add is a local transfer to arbitrary windows, through a smooth-part decomposition, and the replacement of the numerator 4 by m , which turns cubic logarithmic class mass into $(\log X)^3/m$. **Status:** every result is proved *relative to that note*, which is internally proved and has *not* been externally refereed; the same applies to this note. Nothing here bears on the existence of exceptions; this is not a proof of the Erdős–Straus conjecture or of Schinzel’s conjecture.

1. INTRODUCTION

Let $m \geq 2$ be an integer. A positive integer n is *m-representable* if

$$\frac{m}{n} = \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} \tag{1}$$

has a solution in positive integers, not necessarily distinct, and *m-exceptional* otherwise. For $m \leq 3$ every n is representable; for $m \geq 4$ the integer $n = 1$ is exceptional. The Erdős–Straus conjecture is the case $m = 4$ with no exception $n \geq 2$; Sierpiński conjectured the analogue for $5/n$, and Schinzel conjectured that for each $m \geq 4$ there are only finitely many m -exceptional n . For a real interval I , a modulus $q \geq 1$ and $b \in \mathbb{Z}$ put

$$E_m(I) = \#\{n \in I \cap \mathbb{Z}_{\geq 1} : n \text{ } m\text{-exceptional}\},$$

$$E_m(I; q, b) = \#\{n \in I \cap \mathbb{Z}_{\geq 1} : n \equiv b \pmod{q}, n \text{ } m\text{-exceptional}\},$$

and $E_m(N) = E_m((0, N])$.

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Background. Vaughan [V] proved $E_4(N) \ll N \exp\{-c(\log N)^{2/3}\}$, and, according to [PW, abstract], the analogous statement that most n are m -representable for each fixed m . Elsholtz and Tao [ET, §1] cite Vaughan's bound, listing other results as weaker, and Pomerance and Weingartner [PW, p. 2] describe it as the state of the art for $4/n$. Pomerance and Weingartner [PW, Thm. 1.3] made it uniform in the numerator: for $4 \leq m \leq (\log N)^2$,

$$E_m(N) \leq N \exp\{-C_{\text{PW}}(\log N)^{2/3} \varphi(m)^{-1/3}\}. \quad (2)$$

An internal research note [TQ] (“the note” below) proves the bound

$$E_4(N) \ll N \exp\{-c(\log N)^{3/4}\}$$

by an argument that has been internally checked but not externally refereed. All results below are proved relative to [TQ]; Section 2 lists exactly which of its statements are used.

Results. All constants $c, C, \dots$ below are absolute. As in [TQ] they are not effective, because the standard Bombieri–Vinogradov theorem is used.

Theorem A (short intervals, uniform in m). *There are absolute constants $c, C > 0$ such that for every integer $m \geq 4$, every real z and every $H \geq 2$,*

$$E_m((z, z + H]) \leq C H \exp\{-c(\log H)^{3/4} m^{-1/4}\}.$$

In particular $E_m(N) \leq C N \exp\{-c(\log N)^{3/4} m^{-1/4}\}$ for all $N \geq 2$.

The bound depends only on the length of the interval. It contains the theorem of [TQ], which is the case $m = 4$, $z = 0$. The saving tends to infinity if and only if $m = o((\log H)^3)$. For $m \leq \varepsilon(\log H)^3$ with $\varepsilon = \varepsilon(c, C)$ small it is a nontrivial constant factor; for larger m it is no better than the trivial bound $E_m(I) \leq H + 1$.

Theorem B (progressions). *There are absolute $c, C, \alpha > 0$ with the following property. Let $m \geq 4$, let I be an interval of length H , let $q \geq 1$ and $b \in \mathbb{Z}$ with $H/q \geq 2$, put $X_q = \exp\{\alpha(m \log(H/q))^{1/4}\}$, and let q_1 be the X_q -smooth part of q (the product of the prime powers $p^a \parallel q$ with $p \leq X_q$). Then*

$$E_m(I; q, b) \leq C q_1 \frac{H}{q} \exp\{-c(\log(H/q))^{3/4} m^{-1/4}\}.$$

Consequently $E_m(I; q, b) \leq C(H/q) \exp\{-\frac{c}{2}(\log(H/q))^{3/4} m^{-1/4}\}$ whenever $q \leq \exp\{\frac{c}{2}(\log(H/q))^{3/4} m^{-1/4}\}$, and the factor q_1 is absent when every prime factor of q exceeds X_q .

Corollary C (exceptional primes in short intervals). *With c, C as in Theorem A: for $m \geq 4$, $x \geq 3$ and $H \geq 2$ with $(\log H)^{3/4} m^{-1/4} \geq (2/c) \log \log x$,*

$$\#\{p \in (x, x + H] : p \text{ } m\text{-exceptional}\} \leq C \frac{H}{\log x} \exp\{-\frac{c}{2}(\log H)^{3/4} m^{-1/4}\}$$

(p prime). For fixed m this holds for $H \geq \exp\{C_m(\log \log x)^{4/3}\}$ with $C_m = (2/c)^{4/3} m^{1/3}$.

No information on primes at scale x enters: the factor $1/\log x$ is paid for by the saving. Prime-number input is needed only for relative statements (Remark 7.6).

Corollary D (the density transition). *There is an absolute C_D such that for all $m \geq 4$ and all $N \geq 16$ with $\log N \geq C_D m^{1/3} (\log m)^{4/3}$,*

$$\#\{p \in (N/2, N] : p \text{ prime, } m\text{-exceptional}\} \leq \frac{N}{(\log N)^2}.$$

Moreover $E_m(N) = o(N)$ whenever $\log N/m^{1/3} \rightarrow \infty$.

In the other direction, the proof of [PW, Thm. 3.1] shows that most primes in $(N/2, N]$ are m -exceptional when $m^{1/4} \ll \log N \leq (\varphi(m)/(C \log^2 m))^{1/3}$ (Section 8). Pomerance and Weingartner's rigorous bounds placed the transition between $\log N = m^{1/3}$ and $\log N = m^{1/2}$ [PW, p. 2]; their Poisson heuristic [PW, p. 3] predicts $m^{1/3+o(1)}$. Corollary D confirms the density part of that prediction, up to the factor $(\log m)^2(m/\varphi(m))^{1/3}$ in $\log N$. We do not know that the proportion of exceptional primes is monotone in between, and nothing is said about the *largest* m -exceptional integer.

Theorem A and (2) have unrelated constants (ours ineffective), so their comparison is asymptotic. The ratio of the exponents is

$$\frac{L^{3/4} m^{-1/4}}{L^{2/3} \varphi(m)^{-1/3}} = L^{1/12} \frac{\varphi(m)^{1/3}}{m^{1/4}} \gg \frac{(Lm)^{1/12}}{(\log \log 3m)^{1/3}}, \quad L = \log N, \quad (3)$$

by $\varphi(m) \gg m/\log \log 3m$. In the range $4 \leq m \leq L^2$ of (2) its exponent $L^{2/3} \varphi(m)^{-1/3}$ is at least 1, which absorbs the factor C of Theorem A. Hence there is an absolute (ineffective) Λ_0 such that, for $4 \leq m \leq L^2$, Theorem A is stronger than (2) as soon as $(Lm)^{1/12} \geq \Lambda_0 (\log \log 3m)^{1/3}$; in particular for all such m once $N \geq N_0$. Outside this range (2) is not available.

Finally (Section 9) a bound with saving in $\log x$ rather than $\log H$, of the shape $E_m((x, x+H]) \ll H e^{-c(\log x)^{3/4}}$, cannot be proved for windows with $H < e^{c(\log x)^{3/4}}/C$ without proving that there are no exceptions at all beyond some point (Proposition E).

Status and novelty. Every statement labelled PROVED here is proved *relative to* [TQ]: the steps that change are written out in full, and the remaining steps are cited from [TQ] by name. The note [TQ] is internally proved; it has *not* been refereed externally, and neither has this note. The underlying working documents [SH, MN] received three internal hostile reviews, and their repairs are incorporated.

We found no published short-interval or progression version of an exceptional-set bound for (1), for any m (Section 10); the searches were not exhaustive. We make no novelty claim for the *method*, which is that of [TQ], and the forced classes are those of Vaughan, Elsholtz–Tao and Pomerance–Weingartner. Every forced class is an identity valid for all n in the class, and counting congruence classes is translation invariant. An expert would therefore regard the transfer to short intervals as routine for the part of the argument that concerns integers free of small prime factors. What we add is the following: a local replacement for the global semigroup step of [TQ] (Lemma 7.3); the bookkeeping that turns t^3 into t^3/m with absolute constants, with no loss of $\varphi(m)$ or $\log m$ (Sections 3–6); and the consequences above. None of this improves the exponent $3/4$.

Organisation. Section 2 lists what is taken from [TQ] and isolates the single object the rest of the argument needs, a *majorant package* (Definition 2.1). Sections 3–6 construct the package for general m , writing out every step that differs from [TQ].

Section 7 contains the local transfer to arbitrary windows and progressions and proves Theorems A, B and Corollary C. Section 8 proves Corollary D, Section 9 discusses the lower end of the window range, Section 10 the literature, and Section 11 lists open problems.

2. THE BLACK BOX AND THE MAJORANT PACKAGE

2.1. Parameters. We use the parameters of [TQ, (2)] with one change. Fix $\kappa = 1/480$ and the integer D of [TQ, Lemma 3.3]. For $X \geq 3$ put

$$t = \log X, \quad K = \lfloor X^\kappa \rfloor, \quad H_K = K^{10},$$

and use the blocks $I_j = (x_j, \min(2x_j, X)]$, $x_j = 2^j X^{1/2}$, with $z_j = x_j^{1/6}$; for large X one has $z_j \geq X^{1/12} > H_K^2$ [TQ, (3)]. (The note calls H_K simply H ; here H is the length of a window.) The change is the multiplier set:

$$\mathcal{K}_m(K) = \{1 \leq k \leq K : (k, m) = 1\}, \quad L_K = \text{lcm } \mathcal{K}_m(K),$$

in place of $\mathcal{K}(K) = \{k \leq K : k \equiv 1 \pmod{4}\}$. For $\mathcal{J} \subseteq \mathcal{K}_m(K)$ put $h(\mathcal{J}) = \sum_{k \in \mathcal{J}} \varphi(k)/k^2$ and $L_{\mathcal{J}} = \text{lcm } \mathcal{J}$. Throughout, $P_y = \prod_{p \leq y} p$ and $S_y(n) = \mathbf{1}_{(n, P_y)=1}$.

2.2. What is used from the note. External inputs are those of [TQ, §3]: Brun–Titchmarsh in the Montgomery–Vaughan form [MV], the Bombieri–Vinogradov theorem [D, Ch. 28], and Shiu’s theorem [S] (inside [TQ, Lemma 3.3]). The statements of [TQ] enter as follows.

Used verbatim.

- [TQ, Lemma 3.1] (elementary lattice sums); it is stated for every integer $1 \leq k \leq K$ and every $(c, k) = 1$.
- [TQ, Lemma 3.3] (low- ω mass), stated for every $k \leq K$, $(c, k) = 1$.
- [TQ, Lemma 4.1] (second incidence moment). It is stated for $\mathcal{J} \subseteq \mathcal{K}(K)$, but its proof uses only $\mathcal{J} \subseteq [1, K]$, so it holds for $\mathcal{J} \subseteq \mathcal{K}_m(K)$.
- The conditional-independence proof of [TQ, Theorem 6.3] (factorial moments) and [TQ, Lemma 8.1] (the even Bonferroni polynomial Q_r).
- For $m = 4$ only: [TQ, Theorem 8.2] (the assembly) as stated, together with the identity [TQ, Lemma 2.1].

Re-run with 4 replaced by m (each change is written out below): [TQ, Lemma 2.1] (the identity; Lemma 3.1), [TQ, Lemma 2.2] (deduplication and CRT; Lemma 3.3), [TQ, Lemma 3.2] (harmonic sum; replaced by Lemma 4.1), [TQ, Theorem 4.2, Corollary 4.3] (prime slice and fibre masses; Proposition 5.1, Corollary 5.2), [TQ, Lemma 7.1] (void; Lemma 6.1) and [TQ, Theorem 8.2] (assembly; Theorem 6.3).

Replaced. The counting step [TQ, (50)] over $[1, N]$ is replaced by its local form (Lemma 7.1), and the global transfer to all denominators [TQ, §9] (semigroup of exceptional primes and Rankin’s trick) is replaced by a local smooth-part decomposition (Lemma 7.3).

Not used. The simplified route [TQ, §5], the ordered-atom replay after [TQ, Theorem 6.3] with [TQ, Lemmas 6.1, 6.2], and [TQ, §§10–11] (heuristic ceiling, checkpoints). We re-derive only the retained original route of [TQ], with the cutoff $\omega(uv) \leq D \log \log X$ and congestion pruning. (The Cauchy–Schwarz route of [TQ, §5] also appears to transfer, via $\varphi(muv) \geq \varphi(m)\varphi(u)\varphi(v)$, but we have not written it out.)

The substitutions are summarised in Table 1. Where the numerator 4 occurs in [TQ] we checked its occurrence by occurrence. Apart from the unused [TQ, §5], the remark after [TQ, (21)] and [TQ, §§10–11], it appears only in statements that are re-run below: in the identity, in the multiplier condition $k \equiv 1 \pmod{4}$ and [TQ, Lemma 3.2], in the deduplication bound $4uv > K$ of [TQ, Lemma 2.2], in the prime modulus $4uv$ (including the level $q \leq 4x^{1/3}$ in [TQ, (8), (9)] and the main-term step of [TQ, Theorem 4.2]), in the inequality $\varphi(4uv) \geq 2\varphi(u)\varphi(v)$, and in the remark on primes $p \equiv 3 \pmod{4}$ in the proof of [TQ, Lemma 7.1]. The inventory bound $|\mathcal{A}_X| \leq KX^{4/3}$ does not involve the numerator. No step of the density argument (lattice sums, Shiu, congestion, moments, void, Bonferroni) looks at the quadratic character of the forced classes. So the dichotomy between $m \equiv 0 \pmod{4}$ and $m \not\equiv 0 \pmod{4}$ that governs *pointwise* questions for m/n [SX, §15] plays no role here.

object	note [TQ] ($m = 4$)	here (general m)
identity	$k\ell + 1 = 4uvw$	$k\ell + 1 = muvw$
multipliers	$k \leq K, k \equiv 1 \pmod{4}$	$k \leq K, (k, m) = 1$
prime progression	$\ell \equiv -k^{-1} \pmod{4uv}$	$\ell \equiv -k^{-1} \pmod{muv}$
totient bound	$\varphi(4uv) \geq 2\varphi(u)\varphi(v)$	$\varphi(muv) \geq \varphi(m)\varphi(u)\varphi(v)$
harmonic sum	$h = \frac{2}{\pi^2} \log K + O(1)$	$h_m \asymp (\varphi(m)/m) \log K$
fibre mass	$\asymp t^3$	$\asymp s := t^3/m$
selector, depth	$y = Bt^3, r \approx D_B t^3$	$y = \max(2, Bs), r \approx D_B s$
ledger	$e^{O(t^4)}$	$e^{O(ts)} = e^{O(t^4/m)}$
transfer	$[1, N]$, then Rankin	any window, smooth parts

TABLE 1. The substitutions $4 \rightarrow m$ and $[1, N] \rightarrow \text{window}$.

2.3. The majorant package. The transfer of Section 7 uses only the following properties.

Definition 2.1. Let $m \geq 4$, $X \geq 3$, $t = \log X$, $s \geq 1$ and $2 \leq y < X^{1/2}$. An $(m; X, s, y)$ -package with constants (a, C_L) is a function $\nu : \mathbb{Z} \rightarrow \mathbb{R}$ together with a positive integer $\mathcal{M}$ all of whose prime factors are at most X , such that:

- (P1) $\nu(n) \geq 0$ for all n , and $\nu(n) \geq 1$ for every m -exceptional $n \geq 1$ with $(n, P_y) = 1$;
- (P2) $\nu = \sum_i \epsilon_i \mathbf{1}_{a_i \pmod{q_i}}$ is a finite signed combination of residue classes with every $q_i \mid \mathcal{M}$ and $T_{\text{abs}} := \sum_i |\epsilon_i| \leq e^{C_L ts}$;
- (P3) $\mathbb{E}_{\mathcal{M}} \nu \leq 2e^{-as}$, where $\mathbb{E}_{\mathcal{M}}$ is the uniform average over $\mathbb{Z}/\mathcal{M}\mathbb{Z}$ (well defined by (P2)).

Proposition 2.2 (packages exist). *There are absolute constants $a, B, C_L, s_0 > 0$ and X_a such that for every $X \geq X_a$ and every integer $4 \leq m \leq t^3$ with $s := t^3/m \geq s_0$ there is an $(m; X, s, \max(2, Bs))$ -package with constants (a, C_L) .*

For $m = 4$ the note gives such a package directly. Indeed [TQ, Theorem 8.2], with its own family and fixed constants, yields a $(4; X, t^3/4, B_4 t^3)$ -package with constants $(4c_a, 4C_L)$ in the notation of [TQ]: property (P1) holds because every exceptional n avoids every atom [TQ, Lemma 2.2], so $H_X(n) = 0$ and $\nu_X(n) = S_y(n)Q_r(0) = 1$ when $(n, P_y) = 1$; (P2) is [TQ, (44)] with $\mathcal{M}$ from [TQ, (7)]; and (P3) is [TQ, (43)].

Note that [TQ, Theorem 8.2] states the lower bound $\nu_X \geq 1$ only for exceptional primes $n > \max(K, y)$. Its proof, however, uses only the identity, which holds for every positive integer in a forced class, so it gives (P1) as stated. The transfer of Section 7 needs only $y \leq B's + 2$ for some absolute B' , which holds here with $B' = 4B_4$. Thus the case $m = 4$ of Theorems A and B follows from [TQ] and Section 7 alone. For general m , Proposition 2.2 is Theorem 6.3 below.

3. THE ATOM FAMILY FOR GENERAL m

Lemma 3.1 (multiplier identity for m). *Let m, k, ℓ, u, v, w be positive integers with $k\ell + 1 = m uvw$. If $n \geq 1$ and $nv \equiv -u \pmod{k\ell}$, then $s = (nv + u)/(k\ell)$ is a positive integer and*

$$\frac{m}{n} = \frac{1}{suw} + \frac{1}{nsuv} + \frac{1}{nuvw}.$$

In particular every positive integer in the class $-uv^{-1} \pmod{k\ell}$ is m -representable; the inverse exists because $uv \mid k\ell + 1$.

Proof. Over the common denominator $nsuvw$ the numerators sum to $nv + u + s = sk\ell + s = s(k\ell + 1) = m suvw$. $\square$

For $m = 4$ this is [TQ, Lemma 2.1]. The identity was checked in exact rational arithmetic on 4821 random instances with $m \leq 60$ (`scripts/emn_identity.py`); this is a sanity check only.

Definition 3.2 (atom family). Let $\mathcal{A}_X^{(m)}$ be the set of quadruples $A = (k, \ell, u, v)$ with

$$\begin{aligned} k &\in \mathcal{K}_m(K), \quad \ell \in I_j \text{ prime}, \quad H_K < u, v \leq z_j, \\ (u, v) &= (uv, k) = 1, \quad \omega(uv) \leq D \log \log X, \quad m uv \mid k\ell + 1, \end{aligned} \tag{4}$$

with atoms and atom count

$$E_A = \{n : n \equiv -uv^{-1} \pmod{k\ell}\}, \quad H_X(n) = \sum_{A \in \mathcal{A}_X^{(m)}} \mathbf{1}_{E_A}(n).$$

The divisibility $m \mid k\ell + 1$ forces $(k, m) = (\ell, m) = 1$. Unlike [TQ], which uses $k \equiv 1 \pmod{4}$ only, we admit every multiplier coprime to m and let the class $\ell \equiv -k^{-1} \pmod{m}$ vary with k . Restricting to $k \equiv 1 \pmod{m}$ would lose a factor of order $\varphi(m)$ in the multiplier harmonic sum, hence in the fibre mass. As in [TQ], the family is fixed and unpruned, and it does not depend on any residue. We use the probability space of uniform n modulo

$$\mathcal{M} = \text{lcm}(L_K, P_y) \prod_{\substack{X^{1/2} < \ell \leq X \\ \ell \text{ prime}}} \ell, \quad 2 \leq y < X^{1/2}, \tag{5}$$

as in [TQ, (7)]. Every prime factor of $\mathcal{M}$ is at most X .

Lemma 3.3 (deduplication and conditional independence). *For all $X \geq X_0$ (absolute) and every $m \geq 4$:*

- (1) *every m -exceptional integer $n \geq 1$ avoids every atom;*
- (2) *at a fixed ℓ , distinct atoms have distinct projections modulo ℓ ; hence the atoms of a compatible set of distinct atoms have distinct primes ℓ_i , and*

$$\mathbb{P}\left(\bigcap_i E_{A_i}\right) = \frac{\mathbf{1}_{\text{compatible}}}{\text{lcm}(k_1, \dots, k_j) \prod_i \ell_i};$$

- (3) *conditional on any residue $c \pmod{L_K}$ compatible with $S_y = 1$ (and on $S_y = 1$), the coordinates $n \pmod{\ell}$, $X^{1/2} < \ell \leq X$, are independent and uniform, and the atom A is active in the fibre c , i.e. E_A meets it, precisely when $k \mid u + cv$.*

Proof. This is [TQ, Lemma 2.2]; we repeat the proof with 4 replaced by m . Part (1) is Lemma 3.1. If two atoms at the same $\ell \in I_j$ have the same projection modulo ℓ , then $\ell \mid uv' - u'v$; as $|uv' - u'v| < z_j^2 < \ell$, we get $uv' = u'v$, and reducedness gives $(u, v) = (u', v')$. Then $k\ell \equiv k'\ell \equiv -1 \pmod{muv}$ and $(\ell, muv) = 1$ give $k \equiv k' \pmod{muv}$; since $muv > H_K^2 > K$, $k = k'$. So two distinct atoms with the same ℓ are incompatible. All $\ell > X^{1/2} > K$ are coprime to L_K and to P_y , so the Chinese remainder theorem gives the intersection formula and the conditional independence; reduction modulo $k \mid L_K$ gives the activation criterion. $\square$

No upper bound on m is needed here: $(\ell, m) = 1$ is forced by $m \mid k\ell + 1$. Lemma 3.3 was also checked numerically for $m \in \{4, 5, 6, 7, 11, 12\}$ and $\ell \in (1000, 3000)$ (19 446 atoms, no collision; `scripts/emn_identity.py`).

4. THE MULTIPLIER HARMONIC SUM

This section replaces [TQ, Lemma 3.2]. Put

$$S_m(x) = \sum_{\substack{j \leq x \\ (j, m) = 1}} \frac{1}{j}, \quad h_m(K) = h(\mathcal{K}_m(K)) = \sum_{k \in \mathcal{K}_m(K)} \frac{\varphi(k)}{k^2}.$$

Lemma 4.1. *Let $m \geq 4$.*

- (a) *For $x \geq m^2$: $S_m(x) \geq (\varphi(m)/m) \log x$.*
- (b) *For every $K \geq 1$: $h_m(K) \geq (1 - \sum_p p^{-2}) S_m(K) \geq 0.54 S_m(K)$.*
- (c) *For every $K \geq 1$ and every prime $p \nmid m$: $\sum_{k \in \mathcal{K}_m(K), p \nmid k} \varphi(k)/k^2 \leq S_m(K)/p$.*
- (d) *There is an absolute C_h such that $h_m(K) \leq S_m(K) \leq C_h(\varphi(m)/m) \log K$ for $K \geq \max(m, 3)$.*

In particular $0.54(\varphi(m)/m) \log K \leq h_m(K) \leq C_h(\varphi(m)/m) \log K$ for $K \geq m^2$.

Proof. (a) By Möbius inversion over $e \mid m$, $S_m(x) = \sum_{e \mid m} \mu(e) e^{-1} \sum_{i \leq x/e} i^{-1}$. For real $y \geq 1$ one has $\sum_{i \leq y} i^{-1} = \log y + \gamma + \theta_y/y$ with $|\theta_y| \leq 1$, and here $x/e \geq x/m \geq m$. Hence

$$S_m(x) = \frac{\varphi(m)}{m} (\log x + \gamma) - \sum_{e \mid m} \frac{\mu(e) \log e}{e} + R, \quad |R| \leq \frac{2^{\omega(m)}}{x}.$$

Differentiating $f(\sigma) = \sum_{e \mid m} \mu(e) e^{-\sigma} = \prod_{p \mid m} (1 - p^{-\sigma})$ at $\sigma = 1$ gives $-\sum_{e \mid m} \mu(e) \log e / e = f'(1) = (\varphi(m)/m) \sum_{p \mid m} \log p / (p - 1) \geq 0$. Finally $2^{\omega(m)}/x \leq m/m^2 = 1/m < \gamma\varphi(m)/m$, because $\varphi(m) \geq 2$ and $2\gamma > 1$.

(b) Since $\varphi(k)/k = \prod_{p|k} (1 - 1/p) \geq 1 - \sum_{p|k} 1/p$, and $k \in \mathcal{K}_m(K)$ with $p \mid k$ means $k = pj$ with $p \nmid m$ and $j \in \mathcal{K}_m(K/p)$,

$$\begin{aligned} h_m(K) &\geq S_m(K) - \sum_{p \nmid m} \frac{1}{p} \sum_{\substack{k \in \mathcal{K}_m(K) \\ p|k}} \frac{1}{k} \\ &= S_m(K) - \sum_{p \nmid m} \frac{S_m(K/p)}{p^2} \geq \left(1 - \sum_p p^{-2}\right) S_m(K), \end{aligned}$$

and $\sum_p p^{-2} < 0.4523$.

(c) Write $k = pj$ as in (b) and use $\varphi(k)/k^2 \leq 1/k$.

(d) The first inequality is $\varphi(k) \leq k$. Every $j \leq K$ coprime to m has all its prime factors in $\{p \leq K : p \nmid m\}$, so

$$S_m(K) \leq \prod_{\substack{p \leq K \\ p \nmid m}} \left(1 - \frac{1}{p}\right)^{-1} = \prod_{p \leq K} \left(1 - \frac{1}{p}\right)^{-1} \prod_{\substack{p \mid m \\ p \leq K}} \left(1 - \frac{1}{p}\right).$$

For $K \geq m$ every prime factor of m is at most K , so the last product is $\varphi(m)/m$. Mertens' theorem bounds the first by an absolute multiple of $\log K$ for $K \geq 3$. $\square$

A numerical check of (a)–(c) for $m = 4, \dots, 60$ and $m = 210$ up to $K = 10^5$ found no failure, with $S_m(K)/((\varphi(m)/m) \log K) \in [1.05, 1.23]$ (`scripts/emn_hm.py`). For $m = 4$ the note's sub-progression $k \equiv 1 \pmod{4}$ gives $(2/\pi^2) \log K + O(1)$ [TQ, Lemma 3.2], a constant fraction of h_4 .

5. FIBRE MASSES FOR GENERAL m

For $k \geq 1$, $(c, k) = 1$ and $1 \leq H_K < z$ let, as in [TQ, §3],

$$\mathcal{S}(k, c; H_K, z) = \{(u, v) : H_K < u, v \leq z, (u, v) = (v, k) = 1, u \equiv -cv \pmod{k}\}$$

(so $(u, k) = 1$ automatically), and for $\mathcal{J} \subseteq \mathcal{K}_m(K)$ and $(c, L_{\mathcal{J}}) = 1$ let $r_{\mathcal{J}}(u, v; c) = \#\{k \in \mathcal{J} : k \mid u + cv\}$. For $\mathcal{J} \subseteq \mathcal{K}_m(K)$ and a residue c let $f_{c; \mathcal{J}}(\ell)$ be the number of atoms $A \in \mathcal{A}_X^{(m)}$ with $\ell_A = \ell$, $k_A \in \mathcal{J}$ and $k_A \mid u_A + cv_A$.

Proposition 5.1 (pruned prime slice for m). *There are absolute $a_h, A_h > 0$ and X_h such that for every $X \geq X_h$, every integer $4 \leq m \leq t^3$, every $\mathcal{J}$ with $1 \in \mathcal{J} \subseteq \mathcal{K}_m(K)$ and every $(c, L_{\mathcal{J}}) = 1$ the following holds. In each block I_j keep only the triples (k, u, v) with*

$$\begin{aligned} (u, v) &\in \mathcal{S}(k, c; H_K, z_j), \quad k \in \mathcal{J}, \\ \omega(uv) &\leq D \log \log X, \quad r_{\mathcal{J}}(u, v; c) \leq T_X := t^4, \end{aligned} \tag{6}$$

and let $f_{c; \mathcal{J}}^{\text{good}}(\ell)$ be the number of such triples with $muv \mid k\ell + 1$, for $\ell \in I_j$ prime. Then

$$a_h \frac{t^2 h(\mathcal{J})}{\varphi(m)} \leq \sum_{X^{1/2} < \ell \leq X} \frac{f_{c; \mathcal{J}}^{\text{good}}(\ell)}{\ell} \leq \sum_{X^{1/2} < \ell \leq X} \frac{f_{c; \mathcal{J}}(\ell)}{\ell} \leq A_h \frac{t^2 h(\mathcal{J})}{\varphi(m)}.$$

Proof. We follow the proof of [TQ, Theorem 4.2]; the changes are the modulus muv in place of $4uv$ and the resulting factor $1/\varphi(m)$. Each counted triple (k, u, v) together with ℓ is an atom of $\mathcal{A}_X^{(m)}$, active in the fibre c . By Lemma 3.3(2) distinct such

atoms at the same ℓ have distinct classes modulo ℓ , so $f_{c;\mathcal{J}}^{\text{good}} \leq f_{c;\mathcal{J}}$, and both count distinct classes. Fix a full block, $x = x_j$, $z = z_j = x^{1/6}$, $\Lambda = \log(z/H_K) \geq \frac{1}{2} \log z$.

Harmonic mass. Summing [TQ, Lemma 3.3] over $k \in \mathcal{J}$ (its cutoff $D \log \log z$ is at most $D \log \log X$) gives $\sum_{k \in \mathcal{J}} \sum 1/(uv) \geq \frac{1}{8} \Lambda^2 h(\mathcal{J}) \geq \frac{1}{32} (\log z)^2 h(\mathcal{J})$, the inner sum over $(u, v) \in \mathcal{S}(k, c; H_K, z)$ with $\omega(uv) \leq D \log \log X$. The triples removed by $r_{\mathcal{J}} > T_X$ carry harmonic mass at most

$$\sum_{r_{\mathcal{J}}(u, v; c) > T_X} \frac{r_{\mathcal{J}}(u, v; c)}{uv} \leq \frac{1}{t^4} \sum_{\substack{H_K < u, v \leq z \\ (u, v) = 1}} \frac{r_{\mathcal{J}}(u, v; c)^2}{uv} \leq \frac{C_{\text{cg}} (\log z)^2 (1 + \kappa t)^3}{t^4}$$

by [TQ, Lemma 4.1] (whose constant is called C_2 there), which applies to $\mathcal{J} \subseteq [1, K]$. Since $h(\mathcal{J}) \geq 1$ (as $1 \in \mathcal{J}$), the retained harmonic mass is at least $\frac{1}{64} (\log z)^2 h(\mathcal{J})$ for large X .

Main term. A retained triple asks for primes $\ell \equiv -k^{-1} \pmod{q}$, $q = m uv$; this class is reduced because $(k, m) = (k, uv) = 1$. As $m \leq t^3$ and $uv \leq x^{1/3}$, $q \leq t^3 x^{1/3}$. Using $\text{li}(2x) - \text{li}(x) \geq x/\log(2x)$ and $\varphi(m uv) \leq \varphi(m) uv$, the total main-term weight of the retained triples is at least

$$\frac{x}{\varphi(m) \log 2x} \sum_{\text{retained}} \frac{1}{uv} \geq a_1 \frac{x \log x h(\mathcal{J})}{\varphi(m)} \quad (a_1 > 0 \text{ absolute}).$$

Bombieri–Vinogradov. For a fixed q , the quotient $uv = q/m$ is fixed, it has at most $2^{\omega(uv)} \leq (\log X)^{D \log 2}$ ordered coprime splittings (u, v) , and each splitting has at most T_X retained multipliers. So the retained multiplicity of q is $W(q) \leq t^{4+D \log 2}$, exactly as in [TQ, (26)]. Fix $R = D \log 2 + 17$ in [TQ, (9)]. As $x \geq X^{1/2}$ we have $t \leq 2 \log x$, and $q \leq t^3 x^{1/3}$ is below the level $x^{1/2} (\log x)^{-A_R}$ for large X . Grouping the retained triples by q , the total error is therefore at most $(2 \log x)^{4+D \log 2} C_R x (\log x)^{-R} \ll x (\log x)^{-13}$. The main term is at least $a_1 x \log x t^{-3} \gg x (\log x)^{-2}$, by $h(\mathcal{J}) \geq 1$ and $\varphi(m) \leq m \leq t^3$. *This is the only place where the range $m \leq t^3$ is used for the lower bound;* any range $m \leq t^A$ would do with R depending on A . Hence the retained triples have at least $\frac{a_1}{2} x \log x h(\mathcal{J}) / \varphi(m)$ prime incidences $\ell \in (x, 2x]$, all distinct classes, and dividing by $2x$ gives $\sum_{\ell \in (x, 2x]} f_{c;\mathcal{J}}^{\text{good}}(\ell) / \ell \geq \frac{a_1}{4} \log x h(\mathcal{J}) / \varphi(m)$. Summing over the $\asymp t$ full blocks, with $\log x \asymp t$, proves the lower bound.

Upper bound. Drop the pruning and the cutoff. For $(u, v) \in \mathcal{S}(k, c; H_K, z)$, Brun–Titchmarsh [TQ, (8)] with $V = 2x$ and $q = m uv \leq x^{0.34}$, together with $\varphi(m uv) \geq \varphi(m) \varphi(u) \varphi(v)$ (valid since $\varphi(ab) \geq \varphi(a) \varphi(b)$ for all a, b), gives at most $8x / (\varphi(m) \varphi(u) \varphi(v) \log x)$ primes $\ell \in (x, 2x]$. By [TQ, (12)] the sum over $k \in \mathcal{J}$ and (u, v) is $\ll x \log x h(\mathcal{J}) / \varphi(m)$. Divide by x and sum over the blocks, enlarging the last one. $\square$

Corollary 5.2 (uniform fibre masses). *There are absolute $a_u, C_u > 0$ such that for $X \geq X_h$, every integer $4 \leq m \leq t^3$ and every residue $c \pmod{L_K}$, with $\mathcal{J}_c = \{k \in \mathcal{K}_m(K) : (k, c) = 1\}$, $f_c = f_{c;\mathcal{J}_c}$ and $\mu_c = \sum_{\ell} f_c(\ell) / \ell$,*

$$a_h \frac{t^2 h(\mathcal{J}_c)}{\varphi(m)} \leq \mu_c \leq A_h \frac{t^2 h(\mathcal{J}_c)}{\varphi(m)} \leq C_u \frac{t^3}{m}.$$

If $(c, L_K) = 1$, then moreover $\mu_c \geq a_u t^3 / m$.

Proof. As in [TQ, Corollary 4.3]. An atom active in the fibre c has $c \equiv -uv^{-1} \pmod{k}$, a unit, so $k \in \mathcal{J}_c$; thus f_c counts all active atoms, $1 \in \mathcal{J}_c$ and $(c, L_{\mathcal{J}_c}) = 1$, and Proposition 5.1 applies to $\mathcal{J} = \mathcal{J}_c$. For large X , $K \geq t^6 \geq m^2$ and $\kappa t/2 \leq \log K \leq \kappa t$. Lemma 4.1(d) gives $h(\mathcal{J}_c) \leq h_m(K) \leq C_h \kappa t \varphi(m)/m$, hence the upper bound with $C_u = A_h C_h \kappa$. If $(c, L_K) = 1$, then $\mathcal{J}_c = \mathcal{K}_m(K)$ and Lemma 4.1(a),(b) give $h(\mathcal{J}_c) \geq 0.27 \kappa t \varphi(m)/m$, so $a_u = 0.27 \kappa a_h$ works. $\square$

So on reduced fibres the mass is $\asymp t^3/m$ with absolute constants, and on every fibre it is $\ll t^3/m$; for instance $c = 0$ has $\mathcal{J}_0 = \{1\}$ and $\mu_0 \asymp t^2/\varphi(m)$. The factor $1/\varphi(m)$ from the prime progression modulo muv and the factor $\varphi(m)/m$ in h_m cancel to $1/m$, so nothing is lost from $m/\varphi(m)$. A deduplicated toy computation (`scripts/emn_mass2.py`; $\ell \in (10^6, 2 \cdot 10^6]$, $K = 170$, $u, v \in (7, 60]$, two reduced fibres) gave $m\mu_c \in [0.98, 1.13]$ for $m = 4, \dots, 13$, while $\varphi(m)\mu_c$ ranged over $[0.33, 1.04]$. This is EVIDENCE for the scaling only; the corollary does not use it.

6. VOID, MOMENTS AND THE ASSEMBLY FOR GENERAL m

Throughout this section $4 \leq m \leq t^3$ and

$$s = t^3/m \geq 1, \quad y = \max(2, Bs), \quad S_y(n) = \mathbf{1}_{(n, P_y)=1},$$

with an absolute B fixed in Lemma 6.1. For large X , $y \leq \max(2, Bt^3/4) < K < X^{1/2}$.

Lemma 6.1 (conditioned void for m). *There are absolute $a_v, B, s_0 > 0$ and X_v such that for $X \geq X_v$, $4 \leq m \leq t^3$ and $s \geq s_0$,*

$$\mathbb{P}(H_X = 0 \mid S_y = 1) \leq e^{-a_v s}.$$

Proof. We re-run [TQ, Lemma 7.1]. Reveal $c \equiv n \pmod{L_K}$, keeping the condition $S_y = 1$, and put

$$Z(c) = \sum_{\substack{y < p \leq K, \\ p \mid L_K \\ p \mid c}} \frac{1}{p}.$$

For the supported primes $p > y$ the events $p \mid c$ are independent with probabilities $1/p$, also given $S_y = 1$, and the computation [TQ, (39)–(40)] gives $\mathbb{E}(e^{yZ} \mid S_y = 1) \leq \exp\{(e-1)y \sum_{p>y} p^{-2}\} \leq e^{C_Z}$ with C_Z absolute; it uses only $y/p < 1$, so it holds for every $y \geq 2$. Hence $\mathbb{P}(Z > 1/4 \mid S_y = 1) \leq e^{-y/4+C_Z}$.

If $p \mid L_K$ then $p \mid c$ if and only if $p \mid n$; so on $S_y = 1$ no prime $p \leq y$ dividing L_K divides c . A multiplier $k \in \mathcal{K}_m(K) \setminus \mathcal{J}_c$ is divisible by some prime $p \mid (c, L_K)$, and such p satisfy $p > y$ and $p \nmid m$. By Lemma 4.1(b),(c),

$$h(\mathcal{J}_c) \geq h_m(K) - \sum_{\substack{y < p \leq K, \\ p \mid L_K \\ p \mid c}} \frac{S_m(K)}{p} \geq (0.54 - Z(c))S_m(K).$$

On $Z(c) \leq 1/4$, Lemma 4.1(a) (as $K \geq m^2$) and $\log K \geq \kappa t/2$ give $h(\mathcal{J}_c) \geq 0.29(\varphi(m)/m) \log K \geq 0.145 \kappa t \varphi(m)/m$. By Lemma 3.3(3), in the fibre c the event $H_X = 0$ asks that each coordinate $n \bmod \ell$ avoid $f_c(\ell)$ distinct classes, independently

in ℓ . With Corollary 5.2,

$$\begin{aligned} \mathbb{P}(H_X = 0 \mid c, S_y = 1) &= \prod_{\ell} \left(1 - \frac{f_c(\ell)}{\ell}\right) \\ &\leq e^{-\mu_c} \leq \exp\left\{-a_h 0.145 \kappa \frac{t^3}{m}\right\} = e^{-2a_v s}, \end{aligned}$$

with $a_v := 0.0725 \kappa a_h$. Put $B = 8a_v$. For $s \geq 2/B$ we have $y = Bs$, and the bad fibres have probability at most $e^{-2a_v s + C_Z}$. Averaging over c , $\mathbb{P}(H_X = 0 \mid S_y = 1) \leq (1 + e^{C_Z})e^{-2a_v s} \leq e^{-a_v s}$ for $s \geq s_0 := \max(2/B, \log(1 + e^{C_Z})/a_v)$. $\square$

The threshold for Z is the absolute number $1/4$: the union-bound loss $Z \cdot S_m(K)$ is measured against $h_m \geq 0.54 S_m$ in the same units, so no dependence on m enters. In [TQ] the corresponding constant is called η .

Lemma 6.2 (factorial moments). *For $X \geq X_h$, $4 \leq m \leq t^3$, every real $2 \leq y < X^{1/2}$ and every integer $j \geq 1$, $\mathbb{E}((H_X)_j \mid S_y = 1) \leq (C_u s)^j$, where $(h)_j = h(h-1) \cdots (h-j+1)$.*

Proof. The conditional-independence proof of [TQ, Theorem 6.3] applies verbatim. In a fibre c (with $S_y = 1$, which conditions no coordinate $n \bmod \ell$ as $y < X^{1/2}$), Lemma 3.3 shows that H_X is a sum of independent Bernoulli variables with means $f_c(\ell)/\ell$, at most one atom being hit at each ℓ . So $\mathbb{E}((H_X)_j \mid c, S_y = 1) = \sum_{\ell_1, \dots, \ell_j \text{ distinct}} \prod_i f_c(\ell_i)/\ell_i \leq \mu_c^j$, and $\mu_c \leq C_u s$ for every c by Corollary 5.2. $\square$

Theorem 6.3 (assembly for m). *Let a_v, B, s_0 be as in Lemma 6.1 and put $D_B = \max(e^2 C_u, a_v)$ and $C_L = B + 3D_B + 8$. There is an absolute X_a such that for $X \geq X_a$, $4 \leq m \leq t^3$ and $s = t^3/m \geq s_0$, with r the least even integer $\geq D_B s$, the function*

$$\nu_X(n) = S_y(n) Q_r(H_X(n)), \quad Q_r(h) = \sum_{i=0}^r (-1)^i \binom{h}{i},$$

together with $\mathcal{M}$ from (5), is an $(m; X, s, y)$ -package with constants (a_v, C_L) . In particular Proposition 2.2 holds.

Proof. (P1) By [TQ, Lemma 8.1], $Q_r(0) = 1$ and $Q_r(h) \geq 0$ for all integers $h \geq 0$; so $\nu_X \geq 0$. If n is m -exceptional with $(n, P_y) = 1$, then $H_X(n) = 0$ by Lemma 3.3(1), and $\nu_X(n) = 1$.

(P3) By [TQ, Lemma 8.1], $Q_r(h) \leq \mathbf{1}_{h=0} + \binom{h}{r+1}$. By Lemmas 6.1 and 6.2 and $(r+1)! \geq ((r+1)/e)^{r+1}$,

$$\mathbb{E}(Q_r(H_X) \mid S_y = 1) \leq e^{-a_v s} + \left(\frac{e C_u s}{r+1}\right)^{r+1} \leq e^{-a_v s} + e^{-(r+1)} \leq 2e^{-a_v s},$$

since $r+1 \geq D_B s \geq e^2 C_u s$ and $D_B \geq a_v$. Multiply by $\mathbb{P}(S_y = 1) \leq 1$.

(P2) For all integers n , $\binom{H_X(n)}{i} = \sum_{B \subseteq \mathcal{A}_X^{(m)}, |B|=i} \prod_{A \in B} \mathbf{1}_{E_A}(n)$ and $S_y(n) = \sum_{d|P_y} \mu(d) \mathbf{1}_{d|n}$. Expanding, each term has coefficient ± 1 and is empty or a single class modulo $\text{lcm}(d, k_1 \ell_1, \dots, k_i \ell_i)$, which divides $\mathcal{M}$. Choosing $k \leq K$, $\ell \leq X$ and $u, v \leq X^{1/6}$ shows $|\mathcal{A}_X^{(m)}| \leq K X^{4/3} \leq e^{(4/3+\kappa)t}$, as in [TQ, (48)]. Hence, as in [TQ, (49)],

$$\log T_{\text{abs}} \leq \pi(y) \log 2 + \log(r+1) + r(4/3 + \kappa)t.$$

Using $s, t \geq 1$: $\pi(y) \log 2 \leq y \leq Bs + 2 \leq (B + 2)ts$; $\log(r + 1) \leq r \leq D_B s + 2 \leq (D_B + 2)ts$; and $r(4/3 + \kappa)t \leq 2(D_B + 2)ts$. So $\log T_{\text{abs}} \leq C_L ts$. All prime factors of $\mathcal{M}$ are at most X , and $2 \leq y < X^{1/2}$. $\square$

The whole dependence on m is the substitution $t^3 \rightarrow s = t^3/m$ in the saving, the selector and the Bonferroni depth. The cost $(1 + \kappa)t$ of a single atom modulus does not change, which is why the ledger is $e^{O(ts)} = e^{O(t^4/m)}$ and not $e^{O(s)}$.

7. LOCAL TRANSFER TO WINDOWS AND PROGRESSIONS

In [TQ] the majorant is counted on $[1, N]$ in exactly one place, [TQ, (50)], using only $\#\{n \leq N : n \equiv a \pmod{q}\} = N/q + O(1)$ with error at most one. Nothing in that step uses the left endpoint. The prime-number inputs (BV, BT, Shiu) live at the scale $X = e^t$ and serve only to construct the package. The one step of [TQ] that is not local is the passage from primes to all integers in [TQ, §9], through the semigroup generated by exceptional primes and Rankin's trick. We replace it by Lemma 7.3.

Lemma 7.1 (local mean, class by class). *Let $\nu = \sum_i \epsilon_i \mathbf{1}_{a_i \pmod{q_i}}$ be a finite signed combination of residue classes with every $q_i \mid \mathcal{M}$ and $T_{\text{abs}} = \sum_i |\epsilon_i|$. Let $Q \geq 1$, $\beta \in \mathbb{Z}$, and let $I = (z, z + H]$ be any real interval, $H > 0$. Then*

$$\sum_{\substack{n \in I \cap \mathbb{Z} \\ n \equiv \beta \pmod{Q}}} \nu(n) = H \mathbb{E}_{\mathbb{Z}/\text{lcm}(\mathcal{M}, Q)} [\nu \mathbf{1}_{\beta \pmod{Q}}] + \theta T_{\text{abs}}, \quad |\theta| \leq 1.$$

Proof. For each i the set $\{n \equiv a_i \pmod{q_i}\} \cap \{n \equiv \beta \pmod{Q}\}$ is empty or a single class modulo $q'_i = \text{lcm}(q_i, Q)$. In the second case it has density $1/q'_i$ in $\mathbb{Z}/\text{lcm}(\mathcal{M}, Q)$, and every half-open real interval of length H contains $H/q'_i + \theta_i$ of its elements, with $|\theta_i| \leq 1$. Multiply by ϵ_i and sum over i . $\square$

Lemma 7.2 (large-prime moduli are independent). *Let $\nu \geq 0$ be $\mathcal{M}$ -periodic, and let $Q = Q_1 Q_2$ with $(Q_1, Q_2) = (Q_2, \mathcal{M}) = 1$. Then for every β , $\mathbb{E}_{\mathbb{Z}/\text{lcm}(\mathcal{M}, Q)} [\nu \mathbf{1}_{\beta \pmod{Q}}] \leq \mathbb{E}_{\mathcal{M}} \nu / Q_2$.*

Proof. By the Chinese remainder theorem $\mathbb{Z}/\text{lcm}(\mathcal{M}, Q) \cong \mathbb{Z}/\text{lcm}(\mathcal{M}, Q_1) \times \mathbb{Z}/Q_2$. The function ν depends only on the first coordinate, with mean $\mathbb{E}_{\mathcal{M}} \nu$ there, and $\mathbf{1}_{\beta \pmod{Q}} \leq \mathbf{1}_{\beta \pmod{Q_2}}$, which depends only on the second coordinate and has mean $1/Q_2$. Use $\nu \geq 0$. $\square$

Lemma 7.3 (smooth-part decomposition). *Let $m \geq 4$ and $y \geq 2$. Write each integer $n \geq 1$ uniquely as $n = dn'$ with $P^+(d) \leq y$ and $(n', P_y) = 1$, where $P^+(d)$ is the largest prime factor of d ($P^+(1) = 1$).*

- (a) *If n is m -exceptional, then so is n' ; here $n' = 1$ is allowed and is m -exceptional.*
- (b) *If $d > D_0 \geq 1$, then d has a divisor $d' \in (D_0, yD_0]$ with $P^+(d') \leq y$.*
- (c) *For $D_0 \geq 1$, $\sum_{d' > D_0, P^+(d') \leq y} 1/d' \leq D_0^{-1/2} e^{4\sqrt{y}}$.*

Proof. (a) If $m/n' = \sum_i 1/x_i$, then $m/n = \sum_i 1/(dx_i)$. The integer 1 is m -exceptional because three unit fractions sum to at most $3 < m$. (b) Write $d = p_1 \cdots p_r$ with primes $p_i \leq y$ and let i be least with $p_1 \cdots p_i > D_0$; then $d' = p_1 \cdots p_i \leq yD_0$. (c) By Rankin's trick the sum is at most $D_0^{-1/2} \prod_{p \leq y} (1 - p^{-1/2})^{-1}$. By convexity, $-\log(1 - u) \leq 2u$ for $0 \leq u \leq 2^{-1/2}$, and $\sum_{p \leq y} p^{-1/2} \leq \sum_{j \leq y} j^{-1/2} \leq 2\sqrt{y}$. $\square$

Part (a) is the divisor closure used in [TQ, §9]; we apply it to the smooth part only, which keeps everything inside the window.

Proposition 7.4 (transfer). *Let $\nu, \mathcal{M}$ be an $(m; X, s, y)$ -package with constants (a, C_L) , and put $D_0 = e^{as}$. Let I be an interval of length $H > 0$, let $q \geq 1$, $b \in \mathbb{Z}$, and let q_2 be the largest divisor of q all of whose prime factors exceed X . Then*

$$E_m(I; q, b) \leq \frac{H}{q_2} \left(e^{-as/2+4\sqrt{y}} + 2(1+as)e^{-as} \right) + yD_0 + D_0 e^{C_L ts}.$$

Proof. Split the m -exceptional $n \in I$ with $n \equiv b \pmod{q}$ according to their y -smooth part d (Lemma 7.3). Since d is y -smooth and $y < X$, $(d, q_2) = 1$.

Case $d > D_0$. By Lemma 7.3(b), $d' \mid n$ for some y -smooth $d' \in (D_0, yD_0]$. For fixed d' , the $n \in I$ with $d' \mid n$ and $n \equiv b \pmod{q}$ form at most one class modulo $\text{lcm}(d', q)$, a multiple of $d'q_2$; there are at most $H/(d'q_2) + 1$ of them. Summing over d' and using Lemma 7.3(c) gives at most $(H/q_2)D_0^{-1/2}e^{4\sqrt{y}} + yD_0$.

Case $d \leq D_0$. Then $n' = n/d$ lies in I/d , an interval of length H/d ; it is m -exceptional and coprime to P_y , so $\nu(n') \geq 1$ by (P1). The congruence $dn' \equiv b \pmod{q}$ is either insoluble or confines n' to one class β_d modulo $Q_d = q/(d, q)$. As $(d, q_2) = 1$, $Q_d = Q_{d,1}q_2$ with $(Q_{d,1}, q_2) = (q_2, \mathcal{M}) = 1$. By Lemmas 7.1 and 7.2 and (P2), (P3), the number of such n is at most

$$\sum_{\substack{n' \in I/d \\ n' \equiv \beta_d \pmod{Q_d}}} \nu(n') \leq \frac{H}{d} \cdot \frac{2e^{-as}}{q_2} + e^{C_L ts}.$$

Summing over $d \leq D_0$ with $\sum_{d \leq D_0} 1/d \leq 1 + \log D_0 = 1 + as$ gives at most $2(H/q_2)(1+as)e^{-as} + D_0 e^{C_L ts}$. $\square$

If I contains integers $n \leq 0$, they are simply not counted on the left; on the right, $\nu \geq 0$ makes the count over all integers of I/d an upper bound. Every bound depends on I only through H .

Proof of Theorem A and Theorem B. Take a, B, C_L, s_0, X_a from Theorem 6.3, so that $y = \max(2, Bs) \leq Bs + 2$. (For $m = 4$ one may instead use [TQ] directly, as explained after Proposition 2.2.) Fix s_1 with $4\sqrt{Bs} + 2 \leq as/4$ for $s \geq s_1$. For $s \geq \max(s_0, s_1)$, Proposition 7.4 gives

$$E_m(I; q, b) \leq C_1 \frac{H}{q_2} e^{-as/4} + e^{C_2 ts}, \quad (7)$$

with C_1 absolute and $C_2 = C_L + a + B + 3$; for the second term use $y \leq (B+2)s$, $D_0 = e^{as}$ and $s, t \geq 1$.

Let $\mathcal{H} = \log(H/q)$, which is $\log H$ for Theorem A (where $q = 1$). Choose X through

$$t^4 = \frac{m\mathcal{H}}{2C_2}, \quad \text{so that} \quad e^{C_2 ts} = e^{C_2 t^4/m} = (H/q)^{1/2}$$

and

$$s = \frac{t^3}{m} = \left(\frac{\mathcal{H}}{2C_2} \right)^{3/4} m^{-1/4}.$$

This is $X = X_q$ with $\alpha = (2C_2)^{-1/4}$ in Theorem B. The side conditions are: $X \geq X_a$; $m \leq t^3$, i.e. $s \geq 1$; and $s \geq \max(s_0, s_1)$. Since $t^3 = ms \geq 4s$, the first follows from $s \geq (\log X_a)^3/4$. Choose an absolute s_* at least all these thresholds and so large that $s \geq s_*$ implies $as/4 \leq \mathcal{H}/2$; this is possible since $s \leq \mathcal{H}^{3/4}$ once $2C_2 \geq 1$.

If $s \geq s_*$, then $(H/q)^{1/2} \leq (H/q)e^{-as/4}$ and $H/q_2 = q_1 H/q$, so (7) gives $E_m(I; q, b) \leq (C_1 + 1)q_1(H/q)e^{-as/4}$, which is the claim with $c = (a/4)(2C_2)^{-3/4}$. If $s < s_*$, then $\mathcal{H}^{3/4}m^{-1/4} < C_3 := (2C_2)^{3/4}s_*$, and the trivial bound $E_m(I; q, b) \leq H/q + 1 \leq 2H/q \leq 2e^{cC_3}(H/q)e^{-c\mathcal{H}^{3/4}m^{-1/4}}$ suffices. Take $C = \max(C_1 + 1, 2e^{cC_3})$. The two consequences in Theorem B follow from $q_1 \leq q$ and from $q_1 = 1$. $\square$

The parameter X is a real number; the package exists for every real $X \geq X_a$, so the continuous choice of t is legitimate. The proof does not use the semigroup step of [TQ, §9], so for $m = 4$, $I = (0, N]$ it gives a second derivation of the integer statement of [TQ, Theorem 1.1] from its prime-free core [TQ, Theorem 8.2].

Remark 7.5 (smooth moduli and prime moduli). The factor q_1 comes from the crude bound $\mathbf{1}_{\beta}(q) \leq \mathbf{1}_{\beta}(q_2)$ in Lemma 7.2. Conditioning on $n \bmod q_1$ fixes n modulo primes of L_K and P_y , that is, it partially fixes the fibre variable c of Lemma 6.1, and it may do so adversarially: the void bound is an average over c in which bad fibres have probability $e^{-\Omega(y)}$. For example, for $q_1 = \prod_{y < p \leq w} p$ and $b \equiv 0 \pmod{q_1}$ the void lemma's bound gives nothing, since $Z(c) \geq \sum_{y < p \leq w, p|L_K} 1/p$ may exceed $1/4$. We have *not* shown that the conditional mean of ν_X is then large. Such progressions are harmless for the equation itself: every $n \equiv 0 \pmod{q_1}$ is representable as soon as one prime $p \mid q_1$ is. For $m = 4$, every prime modulus q with H/q large is covered without the factor q_1 (with saving constant $c/2$): if $q \leq X_q$ then $q \leq \exp\{\frac{c}{2}(\log(H/q))^{3/4}4^{-1/4}\}$ for large H/q . For general m this absorption fails, because $\log X_q/((\log(H/q))^{3/4}m^{-1/4}) \asymp (m/\log(H/q))^{1/2}$ need not be bounded. Primes q with $\exp\{\frac{c}{2}(\log(H/q))^{3/4}m^{-1/4}\} < q \leq X_q$ are not covered by Theorem B.

Proof of Corollary C. Put $\mathcal{S} = (\log H)^{3/4}m^{-1/4}$. Theorem A bounds the count by $CHe^{-c\mathcal{S}} = CHe^{-c\mathcal{S}/2}e^{-c\mathcal{S}/2}$, and $e^{-c\mathcal{S}/2} \leq 1/\log x$ by hypothesis. $\square$

In the same way Theorem B gives, for primes $p \equiv b \pmod{q}$ in $(x, x + H]$, the bound $C(H/(q \log x)) \exp\{-\frac{c}{4}(\log(H/q))^{3/4}m^{-1/4}\}$ whenever $q \leq \exp\{\frac{c}{2}(\log(H/q))^{3/4}m^{-1/4}\}$ and $(\log(H/q))^{3/4}m^{-1/4} \geq (4/c) \log \log x$. Below the range of Corollary C, Theorem A still saves an unbounded factor over H (for fixed m and, say, $H = (\log x)^A$), but not enough to absorb the normalisation $1/\log x$.

Remark 7.6 (relative statements need prime input). To bound the *proportion* of exceptional primes in $(x, x + H]$ one needs a lower bound for $\pi(x + H) - \pi(x)$. For instance, Baker, Harman and Pintz [BHP] show that $(u - u^{0.525}, u]$ contains $\gg u^{0.525}/\log u$ primes for $u \geq u_0$; we quote this from the literature and did not recheck the exact form of the statement. For $2x^{0.525} \leq H \leq x$, tile $(x, x + H]$ greedily from $u = x + H$ downward by $\lfloor H/(2x^{0.525}) \rfloor \geq 1$ disjoint intervals $(u - u^{0.525}, u]$. This gives $\pi(x + H) - \pi(x) \gg H/\log x$, and with Theorem A the proportion of m -exceptional primes in $(x, x + H]$ is $\ll (\log x) \exp\{-c'(\log x)^{3/4}m^{-1/4}\}$, with $c' = 0.525^{3/4}c$. This conclusion is *CONDITIONAL* on the quoted statement of [BHP], which we could not check against the source. It is $\ll \exp\{-\frac{c'}{2}(\log x)^{3/4}m^{-1/4}\}$ once $(\log x)^{3/4}m^{-1/4} \geq (2/c') \log \log x$. Almost-all versions for shorter H , and versions for primes in progressions, follow in the same way from the corresponding prime-counting results; we have not checked their ranges.

8. THE DENSITY TRANSITION

Proof of Corollary D. Apply Theorem A to $I = (N/2, N]$ and put $L = \log(N/2)$; the count is at most $CN \exp\{-cL^{3/4}m^{-1/4}\}$. Let $f(L) = cL^{3/4}m^{-1/4} - 2\log L$; then $f'(L) > 0$ for $L > (8/(3c))^{4/3}m^{1/3}$. Let $L_0 = Am^{1/3}(\log m)^{4/3}$ with $A \geq (8/(3c))^{4/3}(\log 4)^{-4/3}$, so that f increases on $[L_0, \infty)$. Since $\log \log m \leq \log m$,

$$\begin{aligned} f(L_0) &= cA^{3/4} \log m - 2\log A - \frac{2}{3} \log m - \frac{8}{3} \log \log m \\ &\geq \left(cA^{3/4} - \frac{10}{3} - \frac{2\log A}{\log 4} \right) \log m, \end{aligned}$$

which is at least $\log(2C)$ for A large enough (absolute), as $\log m \geq \log 4$. Put $C_D = 2A$. If $\log N \geq C_D m^{1/3}(\log m)^{4/3}$, then $L \geq \frac{1}{2} \log N \geq L_0$ (as $\log N \geq 2\log 2$), so the count is at most $CNe^{-f(L)}L^{-2} \leq N/(2L^2) \leq N/(\log N)^2$; the last step uses $\sqrt{2}\log(N/2) \geq \log N$, which holds for $N \geq 16$. This is $o(\pi(N) - \pi(N/2))$. For integers, the saving in Theorem A is $(\log N)^{3/4}m^{-1/4} = (\log N/m^{1/3})^{3/4}$. $\square$

The other side. Pomerance and Weingartner [PW, Thm. 3.1] construct many m -exceptional primes. Their proof counts, for primes $p \leq N$, the ones covered by Type I congruences, at most $\ll (N/\varphi(m)) \log^2 N \log^2 m$ when $e^{m^{1/4}} \ll N < e^m$, and the ones covered by Type II congruences, at most $\ll (N/\varphi(m)) \log^2 N \log \log N$ [PW, pp. 6–8]. A prime covered by neither is m -exceptional. The proof is written for one $N = N(m)$, with $\varphi(m)/\log^2 m = C \log^3 N$, but the two counts hold for every N in the stated range. So most primes in $(N/2, N]$ are m -exceptional whenever

$$m^{1/4} \ll \log N \leq (\varphi(m)/(C \log^2 m))^{1/3} \quad (8)$$

with a suitable absolute C . Pomerance and Weingartner state this for $N = N(m)$: “In our proof of Theorem 1.1 we actually show that not only is there one exceptional $n > \exp(m^{1/3-\epsilon})$, but that most prime values of n near this bound are exceptions” [PW, p. 2]. The extension to the whole range (8) is our reading of their proof, checked independently by two internal reviewers against the rendered PDF. Pomerance and Weingartner’s own upper bound (2) gives “most primes representable” only for $\log N \gg m^{1/2}(\log m)^{3/2}$, and they wrote that between $\exp(m^{1/3})$ and $\exp(m^{1/2})$ “there is a transition from ‘usually false’ to ‘usually true’ ” [PW, p. 2].

Combining (8) with Corollary D: most primes near N are m -exceptional for $\log N$ up to $(\varphi(m)/(C \log^2 m))^{1/3}$, and most are m -representable for all $\log N \geq C_D m^{1/3}(\log m)^{4/3}$. The two thresholds differ by a factor $\asymp (\log m)^2(m/\varphi(m))^{1/3}$. We do not know whether the proportion of exceptional primes is monotone in between. The largest m -exceptional integer, which the heuristic of [PW, p. 3] places near $\exp(m^{1/2})$, is not addressed by any of this.

Assessment 8.1 (why the shapes are what they are). Both our argument and Vaughan’s, as reconstructed in [PW, §4], balance a fibre mass μ against a ledger $e^{O(t\mu)}$, with $t\mu \asymp L = \log N$. With $\mu \asymp t^2/\varphi(m)$, the mass of the forced classes of Vaughan and Pomerance–Weingartner, this gives $\mu \asymp L^{2/3}\varphi(m)^{-1/3}$. With $\mu \asymp t^3/m$ (Corollary 5.2) it gives $\mu \asymp L^{3/4}m^{-1/4} = (L/m^{1/3})^{3/4}$, which is at least one exactly when $L \gg m^{1/3}$. Up to bounded and $\log \log$ factors, the mass t^3/m agrees with the heuristic intensity $(\log^3 p/m)(\log \log p)^{O(1)}$ of [PW, p. 3] and with their rigorous Type II first-moment count. Within the architecture of [TQ], the dependence $m^{-1/4}$ is the mass-driven ceiling: the multiplier range $K = X^\kappa$ is already

a power of X , and the range $u, v \leq x^{1/6}$ is what keeps $z_j^2 < \ell$, which distinctness needs. This mirrors the three-quarter ceiling of [TQ, §10].

9. THE LOWER END OF THE WINDOW RANGE

Proposition E. *Let $m \geq 4$. Suppose that for some $x_0, C, c > 0$ and some function $H_0(x) \geq 1$ with $CH_0(x)e^{-c(\log x)^{3/4}} < 1$ for all real $x \geq x_0$ one has*

$$E_m((x, x + H_0(x)]) \leq CH_0(x)e^{-c(\log x)^{3/4}} \quad \text{for all real } x \geq x_0.$$

Then there is no m -exceptional $n > x_0$.

Proof. Each such window contains fewer than one, hence no, exceptional integer. Every integer $n > x_0$ lies in one of them: take $x = n - 1$ if $n - 1 \geq x_0$, and $x = x_0$ otherwise, using $H_0 \geq 1$. $\square$

Thus the target shape $E_m((x, x + H]) \ll He^{-c(\log x)^{3/4}}$ cannot be proved for any $H(x)$ with $CH(x) < e^{c(\log x)^{3/4}}$, for instance for polylogarithmic H , without proving that all large n are m -representable. More generally, a bound CHe^{-S} with $S > \log(CH)$ on a window of length H forces it to be free of exceptions. On the scale $S = c(\log H)^\theta$, any $\theta > 1$, or $\theta = 1$ with $c > 1$, is the full conjecture on long windows, while $\theta \leq 1$ with small c is not.

Define the *window exponent* $\theta_{\text{win}}(m)$ as the supremum of the θ for which $E_m(I) \ll_m |I| \exp\{-c(\log |I|)^\theta\}$ holds for some $c > 0$, uniformly over all intervals I with $|I| \geq 2$.

Proposition 9.1. (a) $\theta_{\text{win}}(m) \geq 3/4$ for every $m \geq 4$. (b) If $\theta_{\text{win}}(m) > 3/4$, then $E_m(N) \ll N \exp\{-c(\log N)^\theta\}$ for some $\theta > 3/4$.

Proof. (a) is Theorem A; (b) is the case $I = (0, N]$. $\square$

So the window problem is at least as hard as the problem over initial segments. The remaining statements of this section are heuristic.

Assessment 9.2 (three regimes). For windows $(x, x + H]$ and fixed m : if $H \geq x^\theta$, Theorem A gives saving $\geq c_\theta(\log x)^{3/4}$; if $e^{c(\log x)^{3/4}} \lesssim H \leq x^{o(1)}$, it gives only $(\log H)^{3/4} = o((\log x)^{3/4})$, and a saving of order $(\log x)^{3/4}$ would need either $\theta_{\text{win}} > 3/4$ or input that depends on the position of the window; if $CH < e^{c(\log x)^{3/4}}$, such a saving is equivalent to having no exception in the window (Proposition E).

Assessment 9.3 (shift-uniform methods). Lemma 7.1 is shift-uniform: it bounds the maximum over z of the number of integers in $(z, z + H]$ that avoid every class of the family. Any argument that uses only “exceptional n avoid the classes of a fixed family $\mathfrak{G}$ ” and the window length is bounded below by this maximum $M_{\mathfrak{G}}(H)$. For such methods, the short-window bound at length H is controlled by the same quantity as the bound over $(0, H]$; it is not equal to it, since $(0, H]$ is one shift among many. Whether $(\log H)^{3/4}$ is optimal for such methods is open. We know of no Jacobsthal- or Maier-type construction of windows of length H containing $\geq He^{-C(\log H)^{3/4}}$ avoiders. Concerning position: the family used at length H has $X = \exp\{\alpha(m \log H)^{1/4}\}$ and period $\mathcal{M} = e^{O(X)}$. If $\mathcal{M} + H \leq x$, which holds for $H \leq \exp\{c(\log \log x)^4/m\}$ with a suitable c , every translate class of the period occurs inside $[x, 2x]$. So for that family the worst window lies in $[x, 2x]$, and the position cannot help a method that uses only this family and the window length.

To go beyond $(\log H)^{3/4}$ in $(x, x + H]$ with $H = x^{o(1)}$, a method would have to use something else: rounding effects of the counts of classes of modulus $q \leq H$ at the specific position, correlations between their intersections, or classes of modulus $> H$, each of which contains at most one point of the window. Nothing here excludes such input.

10. LITERATURE

The following was checked in October 2026 by web searches and by reading the papers where accessible. It is an assessment, not an exhaustive survey.

- Elsholtz and Tao [ET, §1] cite Vaughan’s bound $N \exp\{-c(\log N)^{2/3}\}$ over $n < N$ (“for instance”), and refer to Li Delang [Li] and Yang [Y], among others, for weaker results. Yang’s abstract concerns the initial-segment count; we did not read Li’s or Yang’s papers in full. There is no short-interval or progression statement in [ET].
- Pomerance and Weingartner [PW] describe Vaughan’s 1970 bound as the state of the art for $4/n$ [PW, p. 2], and prove (2) for general m , again over $n \leq N$. The abstract of the cited version (v2) says: “A result of Vaughan is that for each m , most n ’s have m/n representable; we make the dependence on m in this result explicit.” (The first arXiv version said instead that they “generalize a result of Vaughan”.) So, according to [PW], Vaughan’s result already covers each fixed m . We could not read Vaughan’s paper and rely on this secondary statement for the dependence on m . [PW, Thm. 1.3] is the first bound *explicit* in m that we can cite.
- Searches for short-interval, almost-all-short-interval and progression versions of exceptional-set bounds for $4/n$ or m/n , for “Sierpiński $5/n$ exceptional set” and for “ $(\log N)^{3/4}$ ” returned nothing relevant. Verification papers (to 10^{17} and 10^{18}) concern computation, not density.
- Bibliographic status: we read [ET], [PW] (arXiv v2) and [S]. The entries [Li] and [Y] were checked only against the bibliography of [ET]; [V] was not accessible to us; [MV], [D] and [BHP] are standard references that we did not re-read for this note.

We found no published short-interval or progression version, and no $3/4$ -type bound for $m \neq 4$. The bound with exponent $3/4$ is itself only internally established, even for $m = 4$ [TQ]. As far as we can tell, Theorem A is new for every $m \neq 4$, and its uniform dependence on m and Corollary D are new relative to [PW]. The method is that of [TQ] with 4 replaced by m . As explained in the introduction, a specialist would expect the short-interval transfer, and presumably Vaughan’s argument already gives $E_4(I) \ll |I| \exp\{-c(\log |I|)^{2/3}\}$ for intervals. An earlier internal document observed the shift-uniformity of the $3/4$ majorant for large primes [W]; Lemma 7.3 extends this to all integers.

11. OPEN PROBLEMS

Problem 1 (large smooth moduli). Remove the factor q_1 in Theorem B, at least for $(b, q_1) = 1$. For such b , conditioning on $n \equiv b \pmod{q_1}$ forces $p \nmid c$ for every $p \mid q_1$, which only makes the bad event $\{Z(c) > 1/4\}$ less likely. Fixing the coordinates $n \pmod{\ell}$ for $\ell \mid q_1$ changes the fibre product by a factor $\leq \exp\{\sum_{\ell \mid q_1, \ell > X^{1/2}} 2\ell^{-2/3}\} \leq \exp\{2\omega(q_1)X^{-1/3}\}$, using $f_c(\ell) \leq z_j^2 \leq \ell^{1/3}$; this is $1 + o(1)$ when $\omega(q_1) = o(X^{1/3})$,

but not for every q_1 . What is missing is a version of Lemma 6.2 conditional on $n \equiv b \pmod{q_1}$. This is probably routine but has not been attempted. A related question is whether every prime modulus q is covered for general m (Remark 7.5).

Problem 2 (the transition gap). Close the gap $(\log m)^2(m/\varphi(m))^{1/3}$ between (8) and Corollary D. The lower side is the first-moment count of [PW]; the upper side is ours. Is the proportion of m -exceptional primes in $(N/2, N]$ monotone in N in this range? For $m \geq \varepsilon(\log N)^3$ Theorem A is trivial, and in the range (8) most primes are exceptional, so $E_m(N) \gg N/\log N$ there and no saving beyond $\log \log N + O(1)$ is possible. What is the true size of $E_m(N)/N$ in between?

Problem 3 (position-dependent input). Determine $\theta_{\text{win}}(m)$. Equivalently for the methods of Assessment 9.3: are there windows of length H that contain $\geq He^{-C(\log H)^{3/4}}$ integers avoiding the forced classes? If not, what position-dependent input could improve on $(\log H)^{3/4}$ in $(x, x+H]$ for $H = x^{o(1)}$?

Problem 4 (the exponent). Improve the exponent $3/4$ itself, for any m . The ceiling discussion of [TQ, §10] and Assessment 8.1 indicate that this needs a new source of class mass, not a better transfer.

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